

NSL-KDD Dataset

Model	Train Accuracy	Validation Accuracy	Test Accuracy	Precision	Recall	F1-score
Logistic Regression	97.18%	97.11%	84.60%	0.86	0.84	0.84
Random Forest	100.00%	99.91%	86.62%	0.88	0.86	0.86
Decision Tree	100.00%	99.82%	87.20%	0.89	0.87	0.87

Binary Class Classification

Model	Train Accuracy	Validation Accuracy	Test Accuracy	Precision	Recall	F1-score
Logistic Regression	97.08%	99.85%	86.33%	0.96	0.92	0.93
Random Forest	99.99%	99.85%	86.33%	0.96	0.92	0.93
Decision Tree	99.99%	99.84%	82.90%	0.96	0.93	0.94

Multiclass Classification

CIC-IDS Dataset

Model	Train Accuracy	Validation Accuracy	Precision	Recall	F1-score
Logistic Regression	96.74%	96.63%	0.97	0.96	0.97
Random Forest	96.74%	99.99%	1.00	1.00	1.00
Decision Tree	96.74%	99.99%	1.00	1.00	1.00